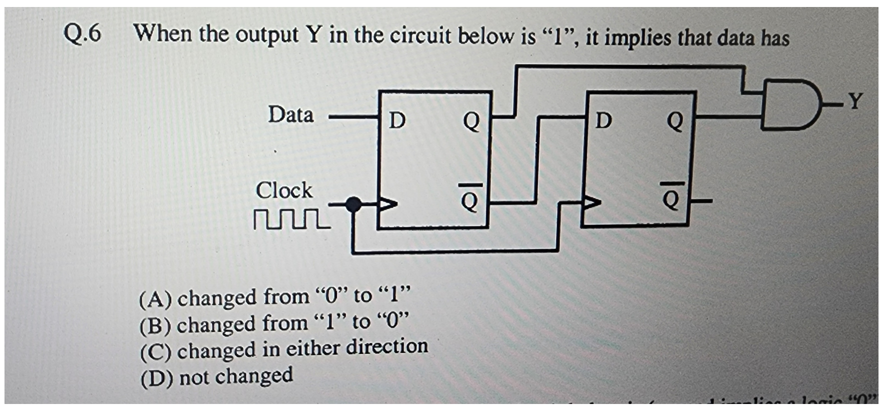
**Name: Surabhi P**
**Id: COMETFWC032**
**Date: 29 August 2025**

# GATE Question solution using Avr-gcc

## Question


Q.6 When the output Y in the circuit below is "1", it implies that data has



(A) changed from "0" to "1"
(B) changed from "1" to "0"
(C) changed in either direction
(D) not changed

Figure 1: Original Q.6 from Exam Paper

## Solution

To determine when output Y in the circuit is "1", analyze the edge-detection logic using two D flip-flops. Y is generated by the AND of the first flip-flop's Q output and the negation of the second flip-flop's Q output:

$$Y = Q_1 \cdot \overline{Q_2}$$

where $Q_1$ is the current data and $Q_2$ is the previous data.

### Truth Table

| $Q_2$ (Previous) | $Q_1$ (Current) | Y |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 0 |
| 1 | 1 | 0 |

### K-Map Minimization

| | $Q_2 = 0$ | $Q_2 = 1$ |
|---|---|---|
| $Q_1 = 0$ | 0 | 0 |
| $Q_1 = 1$ | 1 | 0 |

Thus, the minimized Boolean expression:

$$Y = Q_1 \cdot \overline{Q_2}$$

## Implementation Steps (AVRGCC on Termux)

1. Install Termux and the necessary GCC toolchain for AVR: `pkg install clang avrgcc avrdude`
2. Write the C code below for edge detection
3. Compile your code: `avr-gcc -mmcu=atmega328p -o edge.bin edge.c`
4. Flash to an AVR device using avrdude, or run with a simulator.

## Conclusion

This circuit and corresponding software detect a rising edge on the data line, outputting "1" only when the data transitions from "0" to "1". The function was minimized using a Karnaugh map to $Y = Q_1 \cdot \overline{Q_2}$. The provided C code and hardware description allow implementation with microcontrollers or logic ICs like 7474 in a FSM arrangement for precise edge detection.